Journal of Safety Research xxx (2017) xxx–xxx

Contents lists available at ScienceDirect

Journal of Safety Research

journal homepage: www.elsevier.com/locate/jsr

Special Report from the CDC

Brake reactions of distracted drivers to pedestrian forward collision warning systems

Nils Lubbe *

Toyota Motor Europe NV/SA, Hoge Wei 33, B-1930, Zaventem, Belgium
Autoliv Development AB, Autoliv Research, Wallentinsvägen 22, SE-44783 Vårgårda, Sweden

ARTICLE INFO

Article history:
Received 16 October 2015
Received in revised form 15 October 2016
Accepted 15 February 2017
Available online xxxx

Keywords:
Driver behavior
FCW
Haptic
Head-up display
Reaction time

ABSTRACT

Introduction: Forward Collision Warning (FCW) can be effective in directing driver attention towards a conflict and thereby aid in preventing or mitigating collisions. FCW systems aiming at pedestrian protection have been introduced onto the market, yet an assessment of their safety benefits depends on the accurate modeling of driver reactions when the system is activated. This study contributes by quantifying brake reaction time and brake behavior (deceleration levels and jerk) to compare the effectiveness of an audio-visual warning only, an added haptic brake pulse warning, and an added Head-Up Display in reducing the frequency of collisions with pedestrians. Further, this study provides a detailed data set suited for the design of assessment methods for car-to-pedestrian FCW systems. *Method*: Brake response characteristics were measured for heavily distracted drivers who were subjected to a single FCW event in a high-fidelity driving simulator. The drivers maintained a self-regulated speed of 30 km/h in an urban area, with gaze direction diverted from the forward roadway by a secondary task. *Results*: Collision rates and brake reaction times differed significantly across FCW settings. Brake pulse warnings resulted in the lowest number of collisions and the shortest brake reaction times (mean 0.8 s, SD 0.29 s). Brake jerk and deceleration were independent of warning type. Ninety percent of drivers exceeded a maximum deceleration of 3.6 m/s^2 and a jerk of 5.3 m/s^3. *Conclusions*: Brake pulse warning was the most effective FCW interface for preventing collisions. In addition, this study presents the data required for driver modeling for car-to-pedestrian FCW similar to Euro NCAP's 2015 car-to-car FCW assessment. *Practical applications*: Vehicle manufacturers should consider the introduction of brake pulse warnings to their FCW systems. Euro NCAP could introduce an assessment that quantifies the safety benefits of pedestrian FCW systems and thereby aid the proliferation of effective systems.

© 2017 National Safety Council and Elsevier Ltd. All rights reserved.

UNCORRECTED PROOF

1. Introduction

Driver distraction is a major factor in vehicle-to-pedestrian collisions. Approximately one third of fatal pedestrian accidents in Japan are caused by distracted drivers (ITARDA, 2012), a conclusion further supported by recent in-depth analyses of Japanese naturalistic driving data that found that the primary cause of vehicle-to-pedestrian collisions is the diversion of drivers' attention to something other than the pedestrian at risk (Habibovic, Tivesten, Uchida, Bärgman, & Ljung Aust, 2013).

In situations of potential conflict, Forward Collision Warning (FCW) systems may be effective in directing driver attention towards the conflict and thereby help prevent or mitigate a collision. Automated Emergency Braking (AEB) may also prevent or mitigate collisions by braking the car automatically without driver input in certain conditions. FCW and AEB systems designed with the aim of protecting pedestrians have recently been introduced on the car market (Lubbe & Davidsson, 2015).

FCW systems make use of auditory, visual, and haptic Human Machine Interfaces (HMIs) and must balance driver acceptance with effectiveness in terms of allowing time for a driver to brake the vehicle. The earlier and more intrusive the warning, the more effectively a system can prevent collisions. But at the same time, driver acceptance of such systems decreases, and it has been suggested that drivers are then more likely to ignore the system, turn it off, or refrain from having such systems installed (Lubbe & Davidsson, 2015). This claim is empirically supported by an observational study of car owners' active safety settings: Lane Departure Warning, which was more annoying, was found to be switched off more often than FCW, which was less annoying (IIHS, 2016). A FCW system might therefore act as an Imminent Crash Warning with a rather late activation, aiming for immediate driver reaction, or as a Cautionary Crash Warning with early activation, aiming at directing attention and raising risk awareness, or may incorporate both functions in a triggered approach (Campbell, Richard, Brown, & McCallum, 2007; Naujoks, Grattenthaler, & Neukum, 2012).

* Autoliv Development AB, Autoliv Research, Wallentinsvägen 22, SE-44783 Vårgårda, Sweden.
E-mail address: nils.lubbe@autoliv.com.

http://dx.doi.org/10.1016/j.jsr.2017.02.002
0022-4375/© 2017 National Safety Council and Elsevier Ltd. All rights reserved.

ARTICLE IN PRESS

Assessment methods for evaluating the safety benefit of pedestrian active safety systems including AEB and FCW are under development by Euro NCAP (Euro NCAP, 2015a, 2015b) and NHTSA (Yanagisawa, Swanson, & Najm, 2014). Much attention has been focused on AEB; the automated nature of these systems requires no driver input and simplifies the development of test and evaluation procedures. The Euro NCAP pedestrian active safety assessment, introduced in 2016, includes AEB and FCW evaluation. Euro NCAP has a part score related to AEB effectiveness and a part score related to HMI evaluation. AEB is scored for its ability to reduce vehicle speed prior to a simulated pedestrian impact in four different test scenarios at driving speeds gradually increasing from 20 to 60 km/h. A maximum of 18 scoring points are assigned to each scenario, depending on the speed reduction resulting from testing. FCW is not tested for its ability to reduce impact speed, but is part of the HMI evaluation. FCW is assigned a score of 1 point if operating above 40 km/h and warning at a Time-To-Collision (TTC) of at least 1.2 s (Euro NCAP, 2015b). Further components of the HMI evaluation are an assessment of the system's ease of deactivation (2 points) and night-time operation (1 point). The overall active safety system score is weighted 5:1 for AEB to HMI. As the FCW activation accounts for one fourth of the HMI score, the balance between AEB and FCW is 20:1. Clearly, the relevance of FCW compared to AEB is low in Euro NCAP's assessment. However, the extent to which this reflects the actual safety benefits of these systems has yet to be analyzed.

Driver models, in particular assumptions of driver reaction times and response inputs, are essential in assessing the ability of FCW systems to reduce collisions (Brown, Lee, & McGehee, 2001; McLaughlin, Hankey, & Dingus, 2008). Brake response is more relevant to FCW benefit assessment than steer response as drivers are more likely to respond by braking even in situations where steering would have been more effective in avoiding collision (Adams, 1994). Crucial to the effectiveness of a given FCW system is driver brake reaction time (Brown et al., 2001), brake deceleration and brake jerk. Simply, the earlier and harder the brakes are applied, the higher the speed reduction. For the Euro NCAP safety benefit assessment of car-to-car FCW systems, brakes are applied by a driving robot with brake characteristics specified by the car manufacturer to achieve a deceleration of 4 m/s^2 within 0.2 to 0.4 s (Euro NCAP, 2015c). The specification of these characteristics for pedestrian FCW systems needs to reflect true driver behavior. Such behavior has attracted little attention thus far.

1.1. Brake response for car-to-car FCW

Braking reaction times to FCW have been studied for a variety of FCW systems in simulated car-to-car rear-end collisions (overviews for example in Campbell et al., 2007; Green et al., 2008; Mayer et al., 2011). When warned, drivers must take notice of the warning, decode and understand it, and respond (Ho & Spence, 2008).

Most FCW systems on the market use a combination of visual and auditory warnings (ADAC, 2011). Almost all existing FCW systems in passenger cars use visual displays in the instrument cluster (i.e. near the speedometer) while a few use head-up displays (HUD) projecting the warning signal onto the windscreen area (Wege, Will, & Victor, 2013). Visual displays can readily indicate distinctive threats using icons, but may not be noticed if the driver's view is diverted (Ho, Spence, & Tan, 2005). Head-Up Displays (HUD) can be located closer to the natural gaze direction of the driver to the forward roadway and may be less likely to divert driver attention from the traffic scene compared to displays in the instrument cluster, but it remains unclear whether HUD improves driver reaction compared to instrument cluster warnings (Wege et al., 2013). One aim of this study is to quantify the potential benefits of an added HUD to an audio-visual warning for pedestrian encounters.

Haptic warnings have only been recently introduced as warning interfaces; studies show a potential for improved driver reactions (Ho & Spence, 2008). Some haptic warnings are always noticeable (e.g. seat vibration, or brake pulse) and were shown to be effective in car-to-car rear-end collision driving simulator experiments (Chun, Han, Park, Seo, & Choi, 2012; Mohebbi, Gray, & Tan, 2009; Scott & Gray, 2008). Other haptic warnings depend on the driver being in contact with the warning interface (e.g. pedal vibration), which reduces their overall effectiveness (Campbell et al., 2007).

The findings for the effectiveness of a short brake pulse, as one specific type of haptic warning, are inconclusive. In a car-to-car rear-end collision driving simulator experiment, significantly lower absolute maximum brake deceleration for a brake pulse warning compared to an audio-visual warning was reported. Further, both audio-visual and brake pulse warnings were reported to elicit faster brake reaction than no warning at all by trend (approximately 0.4 s reductions), but the difference was not significant. (Lerner et al., 2011).

Brown et al. (2005) reported a reduced number of potential collisions due to driver braking for brake pulse warnings compared to no warnings in an intersection car-to-car collision warning test track study; however, brake reaction times are not significantly shorter for the brake pulse warning.

Kiefer et al. (1999) reported longer reaction times for brake pulse warnings combined with other warning modalities. Brake pulse–visual warnings lead to longer reaction times than audio-visual warnings alone in a car-to-car forward collision warning track study. In a driving simulator experiment of car-to-car rear-end collisions, Ljung Aust (2014) reported belt–pretensioning–audio and brake pulse–audio warnings having similar effectiveness in significantly reducing reaction times by approximately 0.3 s compared to an audio-visual warning. In a test track experiment of car-to-car rear-end collisions, Kolke, Gauss, and Silvestro (2012) reported reaction times shortened by one-third (approximately 0.3 s, non-significant) when a brake pulse was added to an audio-visual warning.

No studies can be found that measured brake reaction times for brake pulses warnings, separate or when combined with other warnings, in car-to-pedestrian conflicts. Another aim of this study, therefore, is to quantify potential benefits of adding a brake pulse to an audio-visual warning for pedestrian encounters.

1.2. Brake response for pedestrian FCW

Pedestrians are different from cars and the perceived criticality of and driver reactions to car-to-car FCW and pedestrian FCW are not necessarily the same. Pedestrians often enter the driver's field of view from the side (Wisch, Pastor, Zander, & Lorenz, 2012; Yanagisawa et al., 2014), and are a threat smaller in size than a car-to-car collision threat. Event criticality has been shown to influence driver reaction to FCW (Ljung Aust, Engström, & Viström, 2013). Thus, one cannot necessarily assume that the time needed to decode the warning and the driver response is equal for car-to-car and pedestrian threats. Maag, Schneider, Lübbeke, Weisswange, and Goerick (2015) studied drivers approaching vehicle and non-vehicle (including pedestrian) threats and found that driver response is stronger for vehicle threats. It seems thus necessary to study driver behavior in pedestrian encounters in order to obtain a driver model for pedestrian FCW assessment.

The differences observed between warning timings in car-to-car FCW and car-to-pedestrian FCW provides another motivation to study pedestrian FCW. For car-to-car FCW studies, typical ranges are from 2 to 5 s TTC (Jenkins, Stanton, Walker, & Yong, 2007); for example, TTC is 5 s in Mohebbi et al. (2009), 4 s in Chun et al. (2012) and both 3 and 5 s in Scott and Gray (2008). Car-to-pedestrian FCW was measured to activate at a TTC of 1.8 s in a production vehicle (Matsui, Han, & Mizuno, 2011), and thus appears to warn later. Warning time is known to influence driver reaction (e.g. Hirst & Graham, 1997; Scott & Gray, 2008).

Studies of driver reaction times to pedestrian collisions in simulated vehicles are sparse. Straughn, Gray, and Tan (2009) reported steering reaction times for pedestrian FCW. For a warning given at 4 s TTC, haptic

UNCORRECTED PROOF

Please cite this article as: Lubbe, N., Brake reactions of distracted drivers to pedestrian forward collision warning systems, *Journal of Safety Research* (2017), http://dx.doi.org/10.1016/j.jsr.2017.02.002

Fig. 1. Moving base driving simulator dome on motion base (left) and vehicle inside the dome (right). An explanatory video can be accessed at: http://www.youtube.com/embed/uo8lzClpHoA

warnings led to significantly shorter reaction times than auditory warnings. Raudszus et al. (2013) established brake reaction times for mildly distracted drivers in a Driving Simulator study with suddenly-appearing pedestrians. The mild distraction was achieved by "a secondary task that consisted of following the instructions of a navigation system. These instructions were only provided in an audible manner, so that the driver's gaze was not diverted from the street." A pedestrian appeared from behind an obstruction exactly at the time of the warning. Reaction times when receiving an audio-visual warning at 1.8 s or 2.5 s TTC were compared with receiving no warning. FCW had a significant influence on reaction times. Mean brake reaction times reduced from 1.2 s to 0.7 s when warned at 1.8 s TTC and from 1.2 to 1.1 s when warned at 2.5 s TTC. Further, FCW was reported not to affect maximum brake force and the time to reach maximum force significantly. Raudszus et al. (2013) did not report brake jerk or maximum deceleration and did not study the influence of different HMIs on brake response.

1.3. Objectives

This study aims to quantify brake reaction time and brake behavior (deceleration levels and jerk) to provide a detailed data set suited for the design of safety benefit assessment methods for pedestrian FCW systems. Of particular interest to this study were three HMIs: an audio-visual warning, a combination of brake pulse and audio-visual warning, and a novel HUD design highlighting the threat in combination with an audio-visual warning.

The hypotheses for this study were as follows.

- Among FCW systems, Brake pulse and a novel HUD show higher reductions than an audio-visual only HMI.
- Brake reaction time, brake deceleration, and brake jerk are dependent on HMI. Among FCW systems, Brake pulse and novel HUD show shorter reaction times as compared to audio-visual only. Familiarization with the HUD HMI reduces reaction times compared to naive drivers.
- Familiarization with the HUD HMI reduces collision rates and reaction times compared to naive drivers with the same HMI.

Besides testing the hypotheses, driver brake reaction times, brake deceleration and brake jerk, as well as their correlation, are collected and reported here to provide a complete dataset for future modeling of driver reactions to pedestrian FCWs in safety benefit assessments.

2. Methods

Volunteers drove in a simulated urban environment in a moving-base driving simulator. In a single surprise event, when drivers were distracted by a secondary task on a screen mounted on the passenger seat, a simulated pedestrian crossed the road in front of the vehicle on a collision course with the vehicle. Drivers were warned of the imminent threat using different FCW systems. Differences in driver reaction to the FCW were analyzed in a between-subject experimental design.

2.1. Driving simulator

Experiments were conducted in the Toyota driving simulator located at the Toyota Motor Corporation Higashifuji Technical Center in Japan (Fig. 1). The simulator uses an actual vehicle placed on a platform housed within a dome with a diameter of 7.1 m. A 360-degree view is projected on the inside wall of the dome, which is mounted on a 6-degrees-of-freedom motion base. The motion base can move horizontally within a 35 × 20 m range (Adachi et al., 2014).

Volunteers signed a consent form and were then instructed to maintain a speed of 30 km/h and react to traffic as usual before completing a

Fig. 2. Traffic scene at audio-visual warning activation. Left: top view. Right: driver view.

UNCORRECTED PROOF

Please cite this article as: Lubbe, N., Brake reactions of distracted drivers to pedestrian forward collision warning systems, *Journal of Safety Research* (2017), http://dx.doi.org/10.1016/j.jsr.2017.02.002

Table 1
Overview of FCW settings.

	Setting 1 AV	Setting 2 BP	Setting 3 HUD	Setting 4 HUDfam	Setting 5 Control
Warning 1	Audio-visual	Audio-visual	Audio-visual	Audio-visual	None
Activation time	1.8 s TTC	1.8 s TTC	1.8 s TTC	1.8 s TTC	N/a
Warning 2	None	Brake pulse	HUD	HUD	None
Activation time	N/a	1.8 s TTC	2.5 s TTC	2.5 s TTC	N/a
Familiarization with FCW	No	No	No	Yes	N/a

practice course of 3.2 km. The practice course, like the test course, consisted of several intersections in a simulated urban area. Crossing traffic and stationary objects were placed for volunteers to get used to longitudinal and lateral control. After completing the practice course and before driving on the test course, the secondary task was explained to volunteers and demonstrated while they remained in the vehicle. The task, adopted from Forkenbrock et al. (2011), consisted of a series of five random numbers displayed on a screen mounted on the passenger seat to divert the gaze from the forward roadway. The volunteers were instructed to look at the screen, memorize numbers, and repeat them when instructed during the drive. After driving on the test course, volunteers were given a questionnaire asking whether they drove as they would normally do to filter out unnatural driving responses.

The test course had had a total length of 3 km, had no intercepting traffic, and took about 6 min to complete. It ended with the test event. The secondary task was carried out at intersections five times with no simulated pedestrian crossing. On the sixth trial, the actual test was carried out: the pedestrian crossed perpendicular to the vehicle while the driver was engaged in the secondary task. The secondary task was timed to start before pedestrian visibility and last longer than the warning time. The instruction to start the secondary task took place at 3.5 s TTC, while the pedestrian was visible only from 2.7 s TTC. The secondary task screen displayed numbers for 2.5 s in total starting at 3.0 s TTC. The direction of the drivers' gaze was captured by two video cameras directed at the driver's face from the front and side. Tests were excluded if the driver returned the gaze to the forward roadway during the secondary task. Thus, while the driver was looking at the screen mounted on the passenger seat, a simulated pedestrian crossing on a collision course was shown on the dome wall, and the driver warned by one of the three FCW HMIs.

Fig. 2 depicts the traffic scene for a 30 km/h vehicle speed and a 2 m/s pedestrian speed at 1.8 s TTC. A simulated collision with the pedestrian had no influence on vehicle dynamics, and the pedestrian was instantaneously removed from the traffic scene.

2.2. Test event and system specifications

The experiments were conducted with three different HMIs. The same HMI as in Setting 3 was again used in Setting 4, the difference being prior familiarization with the FCW system. A last group, Setting 5, served as control. The settings are summarized in Table 1 and explained in detail below.

- Setting 1 "AV" replicated a Toyota Prius model year 2013 FCW HMI for imminent rear-end car-to-car collisions and included an audio-visual warning issued at TTC 1.8 s. The audio-visual warning HMI consisted of a flashing screen in the instrument cluster near the speedometer displaying the word "brake!" (in Japanese) and a warning sound simultaneously issued at 64 dBA and repeated for 1.8 s (Fig. 3, left).
- Setting 2 "BP" added a short brake pulse, also activated at TTC 1.8 s, to Setting 1. The triangular brake pulse had a duration of approximately 0.2 s and a peak of 3 m/s^2, similar to the pulse of the car-to-car FCW systems of the Lexus LS model year 2014. This brake pulse was intentionally set at a high intensity and clearly noticeable (Fig. 3, center).
- Setting 3 "HUD" issued the same warning as Setting 1 at TTC 1.8 s in a second stage, but pedestrian information was presented in a first stage in the HUD at TTC 2.5 s. Along with this information a single sound at 57 dBA was provided. The HUD showed a framing of the pedestrian indicating the nature and position of the threat (Fig. 3, right).

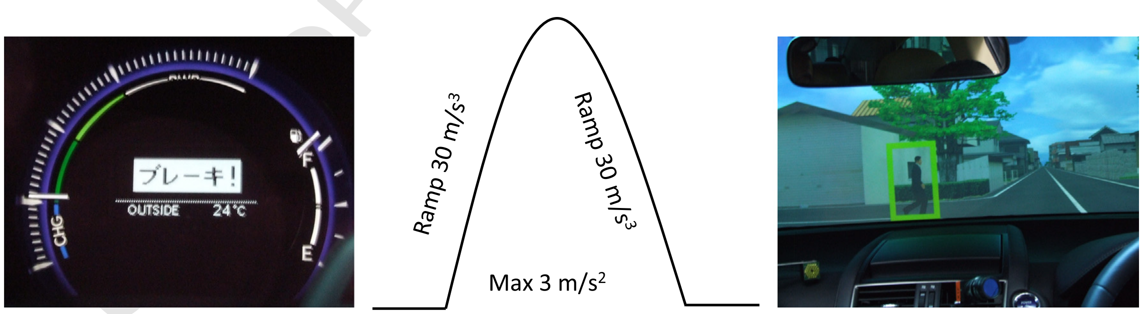

Fig. 3. Setting specifications (from left to right): photo of the instrument cluster warning used in all settings, illustration of the motion base deceleration used to simulate a brake pulse in Setting 2. Photo of the windscreen with the HUD activated as used in Settings 3 and 4.

Table 2
Volunteer characteristics by system: mean (SD).

	Setting 1 AV	Setting 2 BP	Setting 3 HUD	Setting 4 HUDfam	Setting 5 Control
Age [years]	46 (17.5)	53 (14.8)	48 (19.4)	43 (16.4)	52 (18.0)
Yearly mileage [1000 km]	10.8 (8.4)	7.7 (5.4)	10.0 (5.8)	10.5 (6.4)	5.6 (4.5)
Share male [%]	54	46	62	62	50
Number of volunteers	13	13	13	13	6

UNCORRECTED PROOF

Please cite this article as: Lubbe, N., Brake reactions of distracted drivers to pedestrian forward collision warning systems, *Journal of Safety Research* (2017), http://dx.doi.org/10.1016/j.jsr.2017.02.002

ARTICLE IN PRESS

The framing moved together with the pedestrian from the left to the vehicle center.

- Setting 4 "HUDfam" was identical to Setting 3 with respect to the HMI used, but an explanation of system functionality and a demonstration of the HUD, warning and info sound was carried out prior to driving on the test course for these volunteers. Drivers were informed that the system would issue a warning only in cases of imminent collision, and were able to experience warnings produced by the system.
- Setting 5 "Control" included no FCW and served as control.

The HMIs were activated while the drivers were engaged in the secondary task at either 1.8 s TTC (Settings 1 and 2, replicating system activation measured by Matsui et al., 2011) or 2.5 s TTC (Settings 3 and 4, as an earlier cautionary warning).

2.3. Volunteer characteristics

Fifty-eight volunteers approximately representing the gender and age distribution of Japanese driving license holders (Statistics Bureau, 2016) were included in the analysis, matched across the four settings with mean and standard deviation values as presented in Table 2. Gender, age (range: 19 to 78 years) and annual driving distance (range: 10 to 35,000 km) were self-reported and did not differ significantly between groups (age: ANOVA, df = 48, F = 0,73, p = 0.54; mileage: ANOVA, df = 45, F = 0.58, p = 0.63; gender: Fisher–Irwin exact test, p = 0.96).

Volunteers were recruited sequentially to obtain 6 controls and 13 valid tests for each of the four settings, totaling 58 valid tests. In order to obtain these numbers, a total of 101 volunteers were recruited. Of these, 43 were excluded for the following reasons: Tests were incomplete and no data were available due to motion sickness (6 cases); volunteers reported not driving normally during the test (5 cases); volunteers applied the brakes prior to HMI activation (27 cases); and volunteers returned the gaze to the forward roadway during the secondary task (5 cases). The control group was used to verify the distraction task; results are not used in analyses due to its small size.

It should be noted that applying the brake prior to HMI activation does not indicate a failure to engage volunteers in the secondary task for two reasons: First, most volunteers who activated the brake prior to HMI activation in the experiment also did so in the preceding five training situations. Brake application seemed to be triggered by the onset of the secondary task and the volunteers' desire to reduce speed when diverting the gaze from the forward roadway. Second, brake deceleration levels were insufficient to avoid collisions, indicating that these volunteers did not spot the pedestrian and were not trying to avoid a collision prior to HMI activation.

2.4. Measurements

Measurements for the experiment included:

- Collision event (yes or no)
- Brake initiation prior to collision event (yes or no)
- Gaze direction
- Brake reaction time (Time from audio-visual warning at 1.8 s TTC to brake pedal force exceeding 10 N)
- Brake deceleration and jerk (simulated vehicle deceleration)
- Vehicle speed

Table 3
Number of collision events and brake initiations prior to collision.

	Setting 1 AV	Setting 2 BP	Setting 3 HUD	Setting 4 HUDfam
Collision yes	10	1	10	8
Collision no	3	12	3	5
Brake initiation yes	8	13	6	8
Brake initiation no	5	0	7	5

Table 4
Significance levels p in pairwise comparisons using Fisher–Irwin exact test. Collisons (left) and brake initiation (right).

	Collisions				Brake initiation			
	AV	BP	HUD	HUDfam	AV	BP	HUD	HUDfam
AV		<0.01	1	0.67		0.04	0.70	1
BP	<0.01		<0.01	0.01	0.04		<0.01	0.04
HUD	1	<0.01		0.67	0.70	<0.01		0.70
HUDfam	0.67	0.01	0.67		1	0.04	0.70	

Brake pedal force and simulated vehicle deceleration were recorded at 120 Hz and filtered using a rolling average filter with a span of 50.

2.5. Data analysis

For each setting, the number of collisions and the number of collisions avoided were recorded. The Fisher–Irwin exact test (Lachin, 2011) in the SAS software version 9.3 (SAS Institute Inc., Cary, NC) determined association between settings and collision rates. Similarly, the number of brake responses prior to the collision event was compared between settings.

Brake reaction times were measured from the time of warning issue at 1.8 s TTC, common for all settings, until brake initiation, defined as a brake pedal force exceeding 10 N. Differences in settings were analyzed using one-way ANOVA and the Multiple Comparisons function (Tukey's honestly significant difference criterion) in the MATLAB R2013a statistical toolbox. All statistical tests were carried out at a significance level of 5%. All boxplots depict median, first and third quartile; maximum whisker length is 1.5 times the difference between first and third quartile. Data points beyond the whiskers are defined and depicted as outliers using a red + symbol.

Brake responses initiated after having passed the pedestrian cannot be assigned to the FCW with complete certainty; drivers might not have noted the FCW and simulated collision at all. Therefore, a further analysis included only responses from drivers who initiated braking after the FCW but before the simulated collision.

Gaze direction was determined from front and side view video recordings independently by two analysts and averaged. The end of visual engagement (EoVE) was determined as the time from the audio-visual warning (at 1.8 s TTC) to the first frame when either the head or eyes were directed away from the secondary task screen towards the forward roadway. Differences in mean values were analyzed using one-way ANOVA and Multiple Comparison.

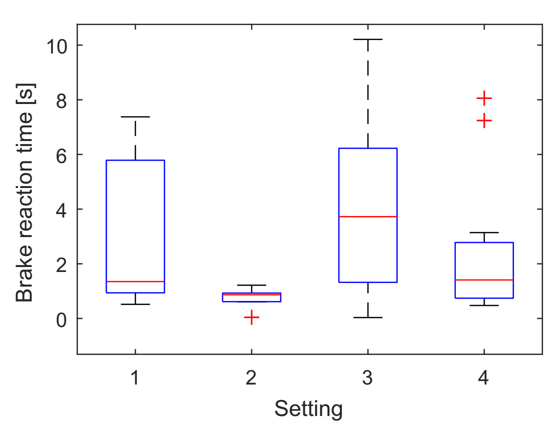

Fig. 4. Brake reaction times.

UNCORRECTED PROOF

Please cite this article as: Lubbe, N., Brake reactions of distracted drivers to pedestrian forward collision warning systems, *Journal of Safety Research* (2017), http://dx.doi.org/10.1016/j.jsr.2017.02.002

Table 5
Brake reaction times of all volunteers (top) and volunteers initiating braking prior to collision (bottom).

	Setting 1 AV	Setting 2 BP	Setting 3 HUD	Setting 4 HUDfam
Mean [s]	2.8 (N = 13)	0.8 (N = 13)	4.1 (N = 13)	2.3 (N = 13)
Standard deviation [s]	2.72	0.29	3.43	2.51
Mean [s]	1.0 (N = 8)	0.8 (N = 13)	1.0 (N = 6)	0.9 (N = 8)
Standard deviation [s]	0.43	0.29	0.59	0.39

Maximum deceleration was calculated as the maximum of simulated vehicle deceleration requested by the driver through pedal activation. Brake jerk was calculated by dividing maximum deceleration by the time needed to reach maximum deceleration from brake initiation.

Vehicle speed was measured at the onset of the audio-visual warning at 1.8 s TTC and at the time of brake initiation, that is when brake pedal force exceeds 10 N.

3. Results

Collisions and collisions avoided per setting are presented in Table 3 and their significance levels in pairwise comparison is reported in Table 4. Differences in collision rates were significant across settings (Fisher–Irwin exact test, $p < 0.01$). The audio-visual plus brake pulse warning (Setting 2) was significantly more effective than any other setting in helping drivers avoid collisions (Table 4, left). Only one collision occurred in this setting. Audio-visual warning and HUD warning without familiarization (Settings 1 and 3) had the highest number of collisions.

In terms of initiated braking, differences were significant across settings ($p < 0.01$). Brake pulse warning (Setting 2) was again significantly more effective than any other setting. Brake pulse warning triggers most brake initiations and is also most effective in preventing collisions; because brake initiation is required for collision avoidance, a strong correlation between these two measures is to be expected.

Brake reaction times (Fig. 4 and Table 5 top) were, on average, shortest for the brake pulse warning (Setting 2, mean 0.8 s, SD 0.29 s). Differences across settings were significant (ANOVA, df = 48, F = 3.81, p = 0.016). Differences in two by two comparison of mean brake reaction time were all not significant except for Setting 2 (BP) versus Setting 3 (HUD) ($p < 0.01$). These reaction times include all volunteer responses, occurring before or after passing the pedestrian collision point.

Considering only those cases where brake response was initiated after the FCW but prior to collision, Setting 2 (BP) had the shortest reaction time as shown in Fig. 5 and Table 5 bottom, which contributes to effective collision avoidance; however, differences across settings were not significant (ANOVA, df = 31, F = 0.81, p = 0.50).

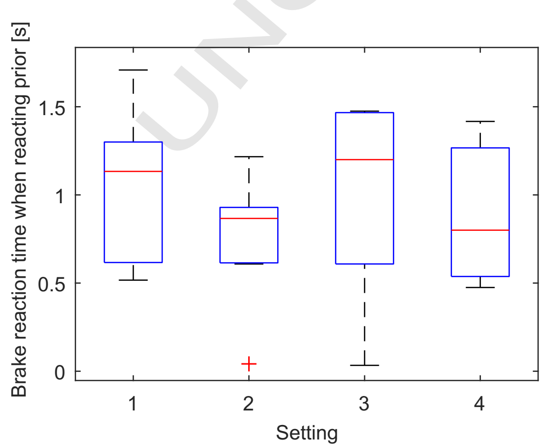

Fig. 5. Brake reaction times of volunteers initiating braking prior to collision.

Similarly, for the analysis of gaze direction, only those cases where braking was initiated before a collision were considered. Again, the mean time from audio-visual warning (at TTC 1.8 s) to EoVE was shortest for the BP warning (Setting 2, see Table 6 and Fig. 6), indicating its ability to alert drivers. The mean time from EoVE to brake initiation was shortest for Setting 4 (Table 6 and Fig. 7). However, differences across settings were statistically not significant (df = 31, F = 0.45, p = 0.72 for time to EoVE; df = 31, F = 0.25, p = 0.86 for time to brake initiation). Notably, Settings 3 and 4, utilizing the novel HUD to indicate the nature and position of the threat, did not decrease the time from EoVE to brake initiation significantly.

For brake response characteristics, again only those cases where braking was initiated before a collision were considered. Differences were not significantly different across settings; they appear only marginally affected by FCW type (max. deceleration with F = 0.046, p = 0.99, Fig. 7, and jerk F = 0.037, p = 0.99, Fig. 8). The mean maximum brake deceleration after the FCW ranged from 7.2 m/s^2 for Setting 1 (AV) to 6.7 m/s^2 for Setting 4 (HUDfam) while brake jerk ranged from 11.3 m/s^3 (Setting 1: AV) to 10.4 m/s^3 (Setting 4: HUDfam) (Fig. 9). Q4 Thus, drivers reacting to a brake pulse warning did not brake harder than drivers experiencing other settings. Brake response characteristics, therefore, do not explain the effectiveness of brake pulse warnings for collision avoidance.

Vehicle speed at 1.8 s TTC (Table 7 and Fig. 10, F = 0.08, p = 0.97) and at brake initiation (Table 7 and Fig. 11, F = 0.74, p = 0.54) did not differ significantly across settings. Volunteers were driving at a speed slightly below the instructed driving speed. The reduction of driving speed for Setting 2 (brake pulse) between audio-visual warning at 1.8 s TTC and driver brake initiation (mean 3 km/h) was due to the mode of warning: the short brake activation, intended as a haptic warning, also reduced driving speed by about 3 km/h. Lower driving speed at brake initiation contributes to collision avoidance.

3.1. Data for driver modeling

Brake reaction time, jerk, and maximum deceleration must be modeled as input for a safety benefit assessment of FCW systems. The proportion of drivers reacting to the FCW prior to collision by initiating braking was significantly different across settings, and should thus be modeled by FCW type. Brake response (max. deceleration and jerk) was only minimally affected by the FCW setting, and does not seem to require modeling by the FCW system.

Brake reaction time correlated neither with maximum brake deceleration (Pearson's linear correlation coefficient $\rho = -0.04$) nor with brake jerk ($\rho = 0.18$), while maximum brake deceleration and brake jerk correlated ($\rho = 0.83$), see Fig. 12.

A multivariate normal distribution and a linear regression were fitted to the correlated data of maximum brake deceleration and jerk (Fig. 13). Contour lines depict percentile values for brake response of the normal distribution, i.e. the "0.05 line" indicates that 5% of brake deceleration and jerk combinations are below and left of the line, and 95% of values lie above and right of the line. The linear regression line depicts representative combinations of brake jerk and brake deceleration; combinations of brake deceleration and brake jerk for a given percentile

UNCORRECTED PROOF

ARTICLE IN PRESS

Table 6
Visual engagement: mean (SD).

	Setting 1 AV (N = 8)	Setting 2 BP (N = 13)	Setting 3 HUD (N = 6)	Setting 4 HUDfam (N = 8)
EoVE [s]	0.65 (0.48)	0.41 (0.04)	0.43 (0.49)	0.53 (0.64)
EoVE to brake initiation [s]	0.39 (0.5)	0.37 (0.2)	0.57 (0.23)	0.36 (0.59)

value can be taken from Fig. 13 as the intercept of regression line and contour line. For example, if a 10-percentile driver model is desired, values of 3.6 m/s^2 deceleration and 5.3 m/s^3 jerk can be read from Fig. 13. For a 90-percentile model, values of 10.8 m/s^2 deceleration and 17.3 m/s^3 jerk can be taken.

4. Discussion

In this study, three different forward collision warning HMIs were tested on independent groups of 13 subjects in a simulated scenario in which a pedestrian crossed the road while the driver's attention was distracted from the forward roadway. The effectiveness of an audio-visual warning only, an added haptic brake pulse warning, and an added Head-Up Display indicating the nature and position of the threat in reducing the frequency of collisions with pedestrians, was studied. A fourth setting comprised the HUD plus an additional familiarization procedure conducted before the test drive. Results showed that brake pulse warnings resulted in the lowest number of collisions and the shortest brake reaction times (mean 0.8 s, SD 0.29 s). Brake jerk and deceleration were independent of warning type. Ninety percent of drivers exceed a maximum deceleration of 3.6 m/s2 and a jerk of 5.3 m/s3. Brake behavior was recorded to provide data needed for driver modeling in safety benefit assessments.

The audio-visual and brake pulse warnings in this study were set to activate at a time-to-collision of 1.8 s, matching a pedestrian FCW system of a production car reported by Matsui et al. (2011). This warning time is in the middle of what Helmer (2014) found to be the optimal range of 1.5 to 2.2 s TTC based on traffic simulation and balancing false activations with injury reduction. Naujoks et al. (2012) found that Cautionary Crash Warnings were appreciated by drivers in a volunteers test as early as 2 to 3 s prior to a conflict evolving into an unavoidable collision. In line with this finding, the HUD was activated at 2.5 s TTC, as an earlier cautionary warning.

Homma, Kikuchi, Wakasugi, Tasaka, and Yoshida (2014) report that only 73% of drivers acknowledging notice of an audio-visual FCW for an imminent but unexpected car-to-car rear-end collision in a track test abandoned the secondary task. The remainder did not respond to the warning, which was attributed to misjudgment of the urgency and warning content. These results are comparable to a reaction rate of 62% of the auditory–visual warnings in Setting 1, which may also include cases where the warning was not noticed at all (Table 3).

The HUD in settings 3 and 4 aimed to assist drivers in locating the threat and inferring the type of threat in line with the recommendation from Campbell et al. (2007) that "warnings should induce an orienting response, where appropriate, causing the driver to look in the direction

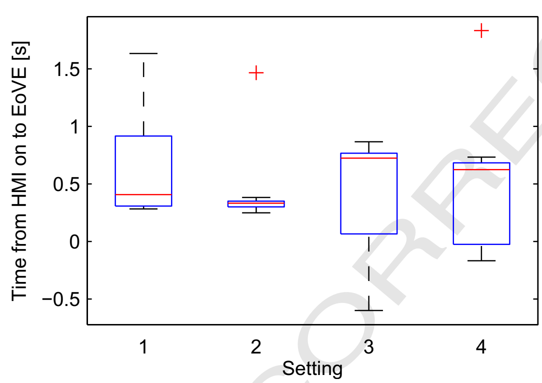

Fig. 6. Time to End of Visual Engagement (EoVE).

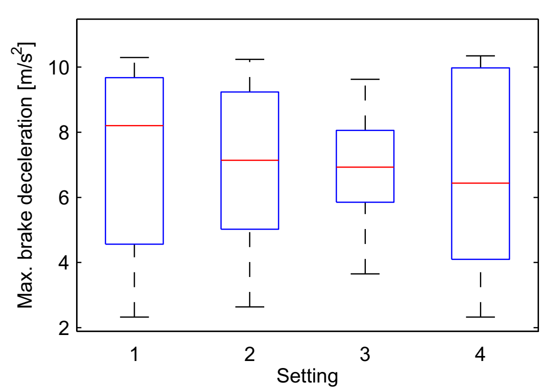

Fig. 8. Maximum vehicle deceleration.

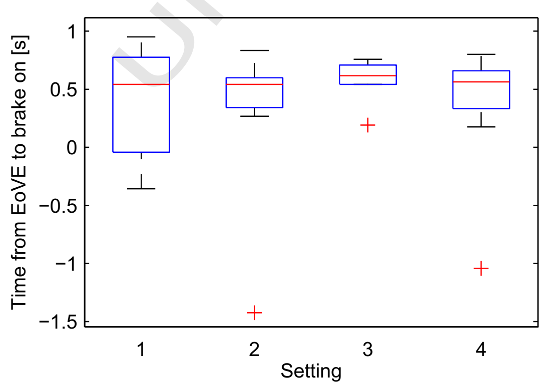

Fig. 7. Time from EoVE to brake initiation'.

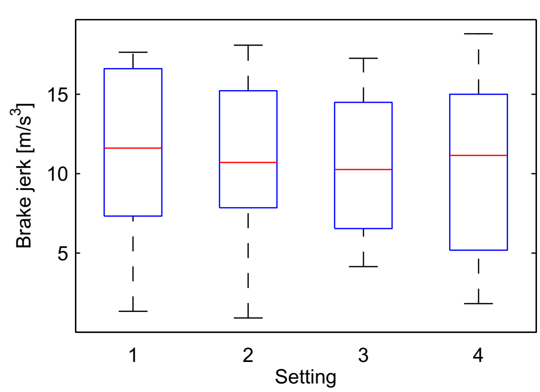

Fig. 9. Vehicle brake jerk.

Please cite this article as: Lubbe, N., Brake reactions of distracted drivers to pedestrian forward collision warning systems, *Journal of Safety Research* (2017), http://dx.doi.org/10.1016/j.jsr.2017.02.002

ARTICLE IN PRESS

Table 7
Vehicle speed: mean (SD).

	Setting 1 AV (N = 8)	Setting 2 BP (N = 13)	Setting 3 HUD (N = 6)	Setting 4 HUDfam (N = 8)
At TTC 1.8 s [km/h]	27.4 (2.9)	27.5 (3.9)	26.5 (4.2)	27.4 (5.2)
At brake initiation [km/h]	26.8 (3.6)	24.4 (3.7)	25.8 (4.3)	26.8 (5.5)

of the Hazard." One might expect a shorter reaction time for the HUD compared to an audio-visual warning (Lees & Lee, 2008). In this study, however, no such benefit of the HUD was observed. There was no effect of the HUD on time to EoVE compared to an audio-visual warning only, and no shortening of the time needed to orient, make a decision, or apply brakes (EoVE to brake on). One can only speculate as to the reasons: The simulated imminent collision situation was simple to comprehend. A single pedestrian in strong contrast to the background crossed the street at a constant speed with no other moving traffic in the vicinity (Fig. 2). It might be that the HUD did not improve an already rapid recognition of the threat.

Familiarization with the HUD prior to the experiment did not result in the expected reduction in collision rates and reaction times compared to naive drivers with the same HMI. One can speculate that the explanation and demonstration given was not sufficient to achieve familiarity with the system, which would imply that drivers might only be able to respond to such warning systems after considerable experience with them. On the other hand, it might also be the case that the warning is so clear that no familiarization is necessary to achieve optimal interaction, that optimal interaction was obtained in this study, and that therefore the HUD is of limited benefit in this context. Further research is necessary.

Previous findings on brake pulse warnings for imminent car-to-car collisions are inconclusive. Some studies might have been using "sub-optimal characteristics of haptic alerts," that is using a haptic warning with unsuitable intensity or timing, so that the conclusions on effectiveness of a haptic warning as such might be influenced by its particular design (Campbell et al., 2007). This study provides evidence that a single triangular brake pulse warning of 0.2 s duration and 3 m/s^2 maximum deceleration was the most effective of the four settings evaluated in this study in alerting distracted drivers to imminent pedestrian collisions. This specific brake pulse warning might be more optimal as a haptic alert than some of those used in previous research. The brake pulse warning was effective through ending distraction and redirecting attention to the road ahead, rather than through leading to quicker brake application once the driver was looking forward. This effect has not been previously reported for brake pulse warnings, and may lead to the conclusion that the reduction of reaction time, in stemming from the redirection of attention, is independent of the collision object. Thus this study provides some evidence that the brake pulse warning for distracted drivers is not only effective for pedestrian encounters, but also likely to be applicable to other types of imminent collisions.

4.1. Strengths and limitations

Driving simulators offer a safe and controlled environment for testing driver response and virtual pedestrian models can move more realistically than in test track studies (Chrysler, Ahmad, & Schwarz, 2015). An advanced moving-base driving simulator is required to realistically provide brake pulse warnings, of which only a few exist. Toyota's high-fidelity moving-base driving simulator was used, which has previously been shown to be able to reproduce driver behavior when encountering pedestrians observed in a test track study (Lubbe & Davidsson, 2015). The use of this particular driving simulator gives strength to the findings.

An advanced driving simulator is, however, costly to run and only a limited number of participants comprise each group. While some interesting findings reported here reach statistical significance, more conclusions could be drawn from a larger sample size.

Further, only demographic information was collected from the participants. Personality characteristics, such as sensation seeking, which may influence the reliability and generalizability of these findings, are not available.

The secondary task proved to be reliable in diverting the drivers' gaze and was simple to verify from video recordings of the drivers. However, the task is not naturalistic, and might give rise to exaggerated distraction levels compared to those encountered in everyday driving.

Volunteers were instructed to drive at 30 km/h. This speed might have been above or below what the volunteers would have felt to be appropriate in that driving situation. The 27 excluded cases of volunteers applying the brakes before warning (reported in Section 2.3) indicate that, for a considerable proportion of volunteers, the driving speed was perceived as too high for proper engagement with the secondary task, hence brake application. Those volunteers who did not apply the brakes prior to the warning might have been comfortable with the driving speed, an assumption supported by the post-experiment statement of driving normally. However, it is also possible that these volunteers also felt uncomfortable with the driving speed but avoided applying the brakes in order to comply with instructions, and did not relate this behavior to un-normal driving. If this were true, the validity of the

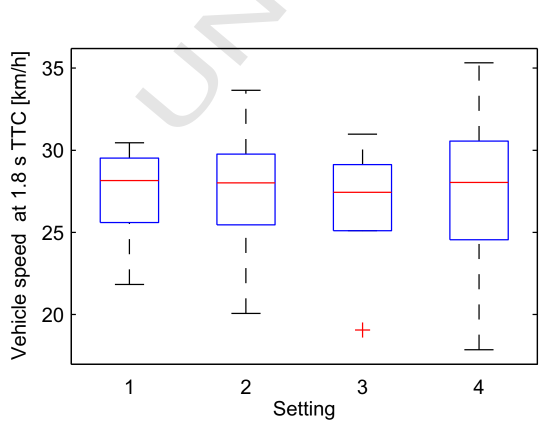

Fig. 10. Vehicle speed at TTC 1.8 s.

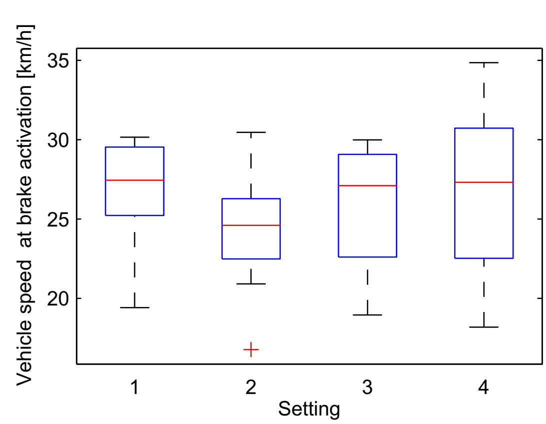

Fig. 11. Vehicle speed at brake initiation.

Please cite this article as: Lubbe, N., Brake reactions of distracted drivers to pedestrian forward collision warning systems, *Journal of Safety Research* (2017), http://dx.doi.org/10.1016/j.jsr.2017.02.002

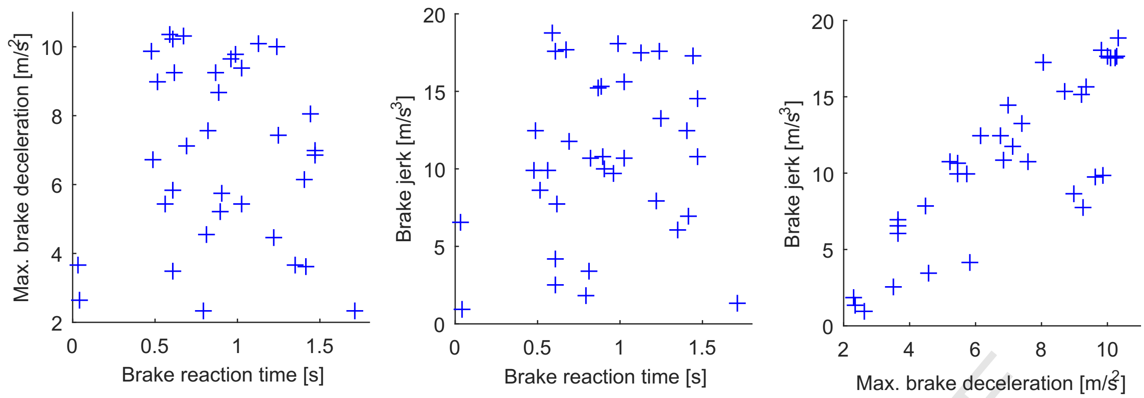

Fig. 12. Correlation between parameters for driver modeling.

results could be questioned. Future research with self-regulated and freely-chosen driving speeds is needed to provide a definitive answer. As brake behavior is known to depend on driving speed, such research would require a larger number of volunteers to compensate for the influence of speed.

Generalization to other types of distraction or to attentive drivers is not straightforward. As mentioned in Section 1.2, Raudszus et al. (2013) reported a mean reaction time of 0.7 s for an audio-visual alert at 1.8 s TTC for mildly distracted drivers with quartile values of approximately 0.4 s and 1.2 s. In contrast, in this study with heavily distracted drivers the mean reaction time was longer (1.0 s) for a similar audio-visual warning. The difference might be explained by assuming that less time needed to redirect attention to the road ahead in the Raudszus et al. study as, unlike in the experiments reported here, driver gaze was not diverted from the forward roadway. Further, Raudszus et al. (2013) reported no cases of drivers not reacting to the warning. The effectiveness of a brake pulse warning may therefore depend on the level of driver distraction: for only mildly distracted drivers, brake pulse may neither increase reaction rate nor reduce reaction time compared to an audio-visual warning, whereas for heavily distracted drivers, adding a brake pulse warning may do both and thus deliver increased safety benefits.

4.2. Application and future research needs

Campbell et al. (2007) noted design issues related to haptic warnings and mentions, among other things, kinematic changes to the driver's position and driver misperceptions and annoyance. These issues are relevant, and can be major obstacles to introducing such systems to the market. For example, false positive activations of this study's high intensity pulse might be perceived as annoying and lead to deactivation of the FCW system. Driver acceptance of brake pulse warnings therefore needs to be studied further.

However, a few systems providing a short brake pulse to alert the driver have been introduced to the market for car-to-car FCW (ADAC, 2011; Hayashi et al., 2013). These systems have not yet been adopted widely and there are presumably challenges there to be addressed; nevertheless, it is encouraging to think that it might be possible to introduce this warning type, shown here to be effective for pedestrian encounters, in the future.

Driver models are needed for assessment schemes such as Euro NCAP's pedestrian AEB rating. In car-to-car AEB assessment, driver reaction to an FCW is modeled by a brake reaction time of 1.2 s and a maximum brake level of 4 m/s^2 (Euro NCAP, 2015c). Such driver reaction for pedestrian FCW might be modeled after data from pedestrian FCW experiments to which this study contributes.

First, the proportion of drivers reacting at all and initiating braking should be modeled separately for each FCW system (Table 3); brake reaction times for these drivers are presented in Fig. 5 and Table 5. Second, brake response in terms of jerk and maximum deceleration could be modeled based on pooled data of volunteer response, independent of FCW system, but correlated to each other as presented in Fig. 13. One only needs to select the percentile value of response relevant for the assessment scheme. For pedestrian FCW safety benefit assessments, data from this study seems not only more relevant than data from car-to-car collision experiments but also enables modeling of correlations and dependencies that could not be modeled using separate studies for each of the model parameters required.

However, the number of subjects per FCW system setting is limited and further studies would be beneficial. Further studies on pedestrian FCW should also quantify the differences in brake response between attentive and distracted drivers, the brake response in more complex scenarios in comparison to the simple one used in this study, the influence of the warning time on brake response, and the driver acceptance of warnings.

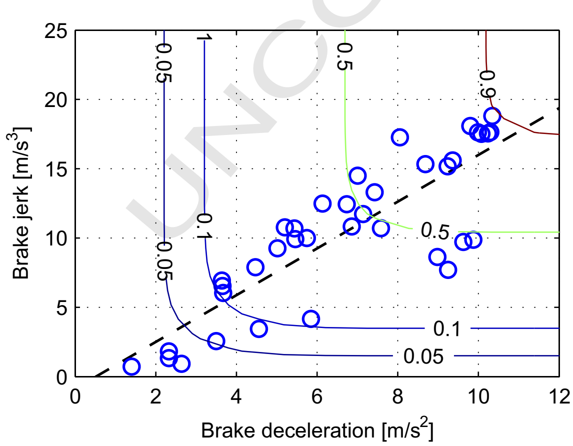

Fig. 13. Brake deceleration and jerk model for FCW response. Blue circles are test values, a dashed line depicts the linear regression, and solid contour lines represent the cumulative normal distribution at the cumulative probability indicated on the line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5. Conclusions

An audio-visual warning with an added brake pulse was most effective in preventing collisions, offering a significant reduction compared to all other FCW settings, namely a novel HUD indicating nature and position of the threat with and without prior familiarization of the volunteers and an audio-visual instrument cluster warning. All drivers initiated braking prior to collision for the brake pulse warning, while only approximately half the drivers did so for the other FCW settings.

UNCORRECTED PROOF

Please cite this article as: Lubbe, N., Brake reactions of distracted drivers to pedestrian forward collision warning systems, *Journal of Safety Research* (2017), http://dx.doi.org/10.1016/j.jsr.2017.02.002

When only those drivers who reacted to the FCW system and started to brake before the collision are considered, brake reaction time, brake deceleration, and brake jerk did not differ significantly across FCW Settings. None of the FCW Settings had a significantly superior performance in ending engagement with the secondary task or in shortening time from starting looking to the forward roadway to brake initiation. The brake pulse warning had the fastest reaction times and lowest driving speeds at brake response, both of which contribute to the effectiveness of collision avoidance.

Data required for a car-to-pedestrian safety benefit assessment similar to that of Euro NCAP's car-to-car scheme is reported. Driver reactions to FCW systems could be modeled based on the presented data reflecting correlations and dependencies in the dataset. However, as the number of subjects per group (n = 13) is relatively low, future research is needed to confirm these findings and expand this preliminary dataset.

References

ADAC (2011). *Comparative test of advanced emergency braking systems. Test report.* Germany: Allgemeiner Deutscher Automobil-Club (Retrieved from: http://www.activetest.eu/pdf/adac_aebs_report_en.pdf).

Adachi, T., Yonekawa, T., Fuwamoto, Y., Ito, S., Iwazaki, K., & Nagiri, S. (2014). Simulator motion sickness evaluation based on eye mark recording during Vestibulo-ocular reflex. *Presented at SAE 2014 World Congress, Detroit, Michigan, USA, paper 2014-01-0441*.

Adams, L. D. (1994). *Review of the literature an obstacle avoidance maneuvers: braking versus steering. Report no. UMTRI-94-19* Transportation Research Inst., Univ. of Michigan.

Brown, T. L., Lee, J. D., & McGehee, D. V. (2001). Human performance models and rear-end collision avoidance algorithms. *Human Factors*, *43*(3), 462–482.

Brown, S. B., Lee, S. E., Perez, M. A., Doerzaph, Z. R., Neale, V. L., & Dingus, T. A. (2005). Effects of haptic brake pulse warnings on driver behavior during an intersection approach. *Proceedings of the Human Factors and Ergonomics Society 49th Annual Meeting*.

Campbell, J. L., Richard, C. M., Brown, J. L., & McCallum, M. (2007). *Crash warning system interfaces: Human factors insights and lessons learned. NHTSA final report no. HS 810 697*.

Chrysler, S. T., Ahmad, O., & Schwarz, C. W. (2015). Creating pedestrian crash scenarios in a driving simulator environment. *Traffic Injury Prevention*, *16*, S12–S17.

Chun, J., Han, S. H., Park, G., Seo, J., & Choi, S. (2012). Evaluation of vibrotactile feedback for forward collision warning on the steering wheel and seatbelt. *International Journal of Industrial Ergonomics*, *42*(5), 443–448.

Euro NCAP (2015a). Assessment protocol — Pedestrian protection, version 8.1 Retrieved from: http://www.euroncap.com/

Euro NCAP (2015b). Test protocol — AEB VRU systems, version 1.0.1 Retrieved from: http://www.euroncap.com/

Euro NCAP (2015c). Test protocol — AEB systems, version 1.1 Retrieved from: http://www.euroncap.com/

Forkenbrock, G., Snyder, A., Heitz, M., Hoover, R. L., O'Harra, B., Vasko, S., & Smith, L. (2011). *A test track protocol for assessing forward collision warning driver–vehicle interface effectiveness. NHTSA final report no. HS 811 501*.

Green, P., Sullivan, J., Tsimhoni, O., Oberholtzer, J., Buonarosa, M. L., Devonshire, J., ... Sayer, J. (2008). *Integrated vehicle-based safety systems (IVBSS): Human factors and driver–vehicle interface (DVI). NHTSA summary report no. HS 810 905*.

Habibovic, A., Tivesten, E., Uchida, N., Bärgman, J., & Ljung Aust, M. (2013). Driver behavior in car-to-pedestrian incidents: an application of the driving reliability and error analysis method (DREAM). *Accident Analysis and Prevention*, *50*, 554–565.

Hayashi, H., Inomata, R., Fujishiro, R., Ouchi, Y., Suzuki, K., & Nanami, T. (2013). Development of pre-crash safety system with pedestrian collision avoidance assist. *Proceedings of Enhanced Safety of Vehicles (ESV) conference, Seoul, South Korea, paper number 13-0271*.

Helmer, T. (2014). *Development of a methodology for the evaluation of active safety using the example of preventive pedestrian protection.* (Dissertation TU Berlin). (available at: http://dx.doi.org/10.14279/depositonce-3973).

Hirst, S., & Graham, R. (1997). The format and presentation of collision warnings. *Ergonomics and safety of intelligent driver interfaces* (pp. 203–219).

Ho, C., & Spence, C. (2008). *The multisensory driver: Implications for ergonomic car interface design*. Ashgate.

Ho, C., Spence, C., & Tan, H. (2005). Warning signals go multisensory. *Proceedings of the 11th Annual Conference on Human–Computer Interaction, 22–27*.

Homma, R., Kikuchi, K., Wakasugi, T., Tasaka, T., & Yoshida, S. (2014). Evaluation of forward-vehicle collision-avoidance systems — How drivers respond to warnings when a hazardous situation is not expected. *Presented at 12th International Symposium on Advanced Vehicle Control, Tokyo, Japan*.

IIHS (2016). *IIHS status report, 51(1)*.

ITARDA (2012). *Accidents resulting in pedestrian fatalities occur most frequently with vehicles proceeding straight ahead.* ITARDA information no. 94 Retrieved from: http://www.itarda.or.jp/itardainfomation/english/info94_e.pdf

Jenkins, D. P., Stanton, N. A., Walker, G. H., & Yong, M. S. (2007). A new approach to designing lateral collision warning systems. *International Journal of Vehicle Design*, *45*, 379–396.

Kiefer, R., LeBlanc, D., Palmer, M., Salinger, J., Deering, R., & Shulman, M. (1999). *Development and validation of functional definitions and evaluation procedures for collision warning/avoidance systems. NHTSA final report HT 808 964*.

Kolke, R., Gauss, C., & Silvestro, D. (2012). Reduzierung von Unfällen durch Notbremssysteme bei Pkw. *Presented at 8th ADAC/BASt-Symposium, Baden-Baden, Germany* (Retrieved from: http://www.adac.de/_mmm/pdf/fv_8adac_bast_symposium_kolke_149074.pdf).

Lachin, J. M. (2011). *Biostatistical methods: The assessment of relative risk* (2nd ed.). Wiley.

Lees, M. N., & Lee, J. D. (2008). Enhancing safety by augmenting information acquisition in the driving environment. In C. Castro (Ed.), *Human factors of visual and cognitive performance in driving*. Taylor & Francis (2008).

Lerner, N., Jenness, J., Robinson, E., Brown, T., Baldwin, C., & Llaneras, R. (2011). *Crash warning interface metrics: Final report. NHTSA final report no. HS 811 470a* .

Ljung Aust, M. (2014). Effects of haptic versus visual modalities when combined with sound in forward collision warnings. *Proceedings of driving simulation conference 2014, Paris, France*.

Ljung Aust, M., Engström, J., & Viström, M. (2013). Effects of forward collision warning and repeated event exposure on emergency braking. *Transportation Research Part F: Traffic Psychology and Behaviour*, *18*, 34–46.

Lubbe, N., & Davidsson, J. (2015). Drivers' comfort boundaries in pedestrian crossings: A study in driver braking characteristics as a function of pedestrian walking speed. *Safety Science*, *75*, 100–106.

Maag, C., Schneider, N., Lübbeke, T., Weisswange, T. H., & Goerick, C. (2015). Car gestures — Advisory warning using additional steering wheel angles. *Accident Analysis and Prevention*, *83*, 143–153.

Matsui, Y., Han, Y., & Mizuno, K. (2011). Performance of collision damage mitigation braking systems and their effects on human injury in the event of car-to-pedestrian accidents. *Stapp Car Crash Journal*, *55*, 461–478.

Mayer, C., Tsuchida, J., Ranovona, M., Chin, E., McCarthy, M., Gelau, C., & Dukic, T. (2011). *Experimental design for an evaluation of behavioural aspects. ASSESS report D3.1.* (Retrieved from: http://www.assess-project.eu/downloadables/Public%20Deliverables/ASSESS%20D3.1-FINAL%20-%20Experimental%20Design-PUBLIC-2010.03.31.pdf).

McLaughlin, S. B., Hankey, J. M., & Dingus, T. A. (2008). A method for evaluating collision avoidance systems using naturalistic driving data. *Accident Analysis and Prevention*, *40*, 8–16.

Mohebbi, R., Gray, R., & Tan, H. Z. (2009). Driver reaction time to tactile and auditory rear-end collision warnings while talking on a cell phone. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, *51*(1), 102–110.

Naujoks, F., Grattenthaler, H., & Neukum, A. (2012). Zeitliche Gestaltung effektiver Fahrerinformationen zur Kollisionsvermeidung auf der Basis kooperativer Perzeption. In K. Dietmayer, K. Bengler, B. Färber, M. Maurer, C. Stiller, & H. Winner (Eds.), *8. Workshop Fahrerassistenzsysteme — FAS 2012 (S. 107–117)*.

Raudszus, D., Ranovona, M., Geronimi, S., Kunert, M., Schubert, E., & Schaller, T. (2013). *Report on driver and pedestrian reaction model. AsPeCSS report D2.4.* (Retrieved from: http://www.aspecss-project.eu/userdata/file/Public%20deliverables/ASPECSS-D2.4-FINAL-Driver%20Model-2013-06-27.pdf).

Scott, J. J., & Gray, R. (2008). Comparison of visual, auditory and tactile warnings for rear-end collision prevention in driving. *Human Factors*, *50*, 264–275.

Statistics Bureau, Ministry of Internal Affairs and Communications, Japan (2016i). *Japan statistical yearbook 2016.* (Retrieved from: http://www.stat.go.jp/english/data/nenkan/index.htm).

Straughn, S. M., Gray, R., & Tan, H. Z. (2009). To go or not to go: Stimulus–response compatibility for tactile and auditory pedestrian collision warnings. *IEEE Transactions on Haptics*, *2*(2), 111–117.

Wege, C., Will, S., & Victor, T. (2013). Eye movement and brake reactions to real-world brake-capacity forward collision warnings — a naturalistic driving study. *Accident Analysis and Prevention*, *58*, 259–270.

Wisch, M., Pastor, C., Zander, O., & Lorenz, B. (2012). Reasons why passive pedestrian protection remains an issue for the future. *Presented at: Crash.Tech conference 2012, Munich, Germany*.

Yanagisawa, M., Swanson, E., & Najm, W. G. (2014). *Target crashes and safety benefits estimation methodology for pedestrian crash avoidance/mitigation systems. Report no. DOT HS 811 998*.

Nils Lubbe received a PhD degree in Machine and Vehicle Systems from Chalmers University of Technology, Gothenburg, Sweden, in 2015. Prior to his current role as research specialist and global real-life safety analysis lead at Autoliv Research he worked in safety evaluation and safety research at Toyota Motor Europe focusing on the reduction of road traffic injuries of pedestrians.

Please cite this article as: Lubbe, N., Brake reactions of distracted drivers to pedestrian forward collision warning systems, *Journal of Safety Research* (2017), http://dx.doi.org/10.1016/j.jsr.2017.02.002